A Multi-Agent, Regime-Aware Systematic Trading Engine

Signal fusion, LLM strategy synthesis, a CVaR meta-agent, and the NEXUS control plane

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Abstract. This note describes the firm's systematic trading engine: ~22 containerised services across **FX, commodities, futures, crypto and options** in which specialist agents extract signal, a weighted-ensemble fusion layer forms a directional view, a code-generating LLM synthesises and walk-forward-validates strategies, and a four-component **regime-aware meta-agent (A3f-cvar)** scales gross exposure by a Monte-Carlo CVaR target before risk-gated execution. The whole estate is governed through **NEXUS**, an operational control plane that enforces a human approval gate and an event-sourced audit trail. On a 20-year proxy backtest the meta-overlay raises out-of-sample Sharpe to **0.895** (above in-sample) and cuts maximum drawdown by **41%**; in a Global-Financial-Crisis stress test trained with zero crisis data it returns **+1.74%** against the S&P's −17.82%. The model is documented to U.S. Federal Reserve SR 11-7 standard.

1. Architecture & signals

The engine runs ~22 Docker services over a **Redis pub/sub backbone** (no direct coupling), persisting to PostgreSQL, with a FastAPI + React operations layer. Six specialist agents publish per-asset: **technical** (RSI, MACD, EMA, Bollinger, ATR, VIX-regime), **macro** (rate differentials / carry + economic-calendar blackout gating), **news/sentiment** (65+ articles, 7 feeds), **fundamental positioning** (CFTC Commitment-of-Traders), and dedicated **risk-sentiment** and **commodity** agents.

2. Fusion & strategy synthesis

A fusion layer combines the families as a weighted ensemble (technical 40 / macro 25 / fundamental 20 / news 15) with an LLM arbitration step over conflicts. A strategy agent then **generates executable LEAN strategy code**, backtested on QuantConnect across a library — trend-following, mean-reversion, momentum, EMA-crossover, London-breakout, statistical-arbitrage. Promotion requires a walk-forward gate (in-sample Calmar ≥ 0.25, out-of-sample ≥ 0.12, OOS/IS ≥ 0.50); validated strategies are PROTECTED, degenerate ones auto-pruned.

3. The regime-aware meta-agent (A3f-cvar) `in production`

Four composable, individually-tested components output a single gross-exposure multiplier in [0.05, 1.0]:

Component	Method	Evidence
Regime detection	3-state Gaussian HMM, NBER-informed prior	100% NBER crisis precision
Anomaly alarm	One-Class SVM (RBF, ν=0.05)	9.5× recession-rate lift on flagged months
Strategy weighting	Normal-Normal Bayesian ensemble + Thompson sampling	tilts to defensives in crisis cell
Exposure sizing	Monte-Carlo CVaR ($N=10^4$, $\alpha=0.95$)	scalar ≈0.99 calm → ≈0.40 crisis

$$\text{maximise } s \in [0.05,\ 1.0] \text{ s.t. } \mathrm{CVaR}_{0.95}(\ s \cdot \tilde{R}_{\mathrm{regime}}\ ) \geq -L_{\mathrm{target}}$$

4. Risk model & execution

Hard gates precede every order — positive Sharpe, win-rate ≥ 50%, position < 10% NAV, correlation guard, and a portfolio **circuit-breaker halting all trading at −15% drawdown**. Sizing is ATR-based (stops 2.0× ATR, targets 3.0× ATR). Execution routes to OANDA (FX/commodities) and Alpaca (equities), with crypto and an **options engine** (SVI/SABR vol surface, py_vollib Greeks; structures held to net-delta ±30% NAV, net-theta ≥ −0.1% NAV/day). A position manager runs stop / take-profit / time exits each minute.

5. Backtest validation

Configuration	Sharpe	Max DD	$CVaR_{95}$
Equal-weight baseline	0.831	−19.34%	−5.50%
Meta-agent (in-sample)	0.834	−11.41%	−3.17%
Meta-agent (out-of-sample WF)	**0.895**	−13.14%	−4.26%

The decisive signal is that **out-of-sample Sharpe (0.895) exceeds in-sample (0.834)** — the opposite of overfitting — over 182 walk-forward decisions, with a 41% drawdown reduction versus equal-weight. In a GFC stress test whose training window contains no crisis, the overlay returns **+1.74%** (max DD −7.49%) versus the S&P's −17.82%, beating long-equity by 19.5 points. 83 automated tests pass.

6. Governance — NEXUS & SR 11-7

Every consequential action flows through **NEXUS**, the firm's operational control plane — a single authenticated surface (the "air-traffic control" for the agent estate) with an **approval inbox** (risk-tiered, SLA-timed), a live React-Flow **agent graph**, per-agent cognitive-trace terminals, and an event-sourced audit trail. The trading model itself is documented in a 949-line paper formatted to the Federal Reserve's **SR 11-7** model-risk guidance (three lines of defence, ongoing-monitoring cadence, recalibration and decommission triggers, full academic citation).

7. Frontier roadmap

- **Full autonomy under guardrails** — the meta-overlay sizes live capital end-to-end, with NEXUS retaining a hard human kill-switch and circuit breaker.
- **Quantum-enhanced signal & risk** — the quantum ensemble (separate write-up) feeding the meta-agent, and quantum amplitude estimation accelerating Monte-Carlo CVaR.
- **Cross-asset options & vol-surface trading at scale**, with regime-conditional CVaR re-calibrated for non-linear payoffs.

Performance figures are from proxy backtests and paper/shadow trading; code is private (NDA) and can be walked through in interview.
`in production` = built & running; `frontier` = next-generation programme.